

Justin Lee

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EDUCATION

Stony Brook University

Master of Science in Applied Math (Focus in AI/ML)

Stony Brook, NY

Aug 2026 – May 2027

Stony Brook University | GPA: 3.50/4.0

Bachelor of Science in Computer Science ; Applied Math

Stony Brook, NY

Aug 2023 – Dec 2026

- **Coursework:** Data Structures, Advanced Algorithms, Operating Systems, Computer Networks, Artificial Intelligence, Probability Theory, Statistical Learning, Fundamentals of LLMs, Software Engineering

EXPERIENCE

Incoming Software Engineer Intern

Betterment

June 2026 – August 2026

New York City, NY

Software Engineer Intern

Acadia Analytics

May 2025 – August 2025

Cambridge, Massachusetts

- Designed Flask and Express services for forecasting workflows, cutting retraining time by **30%**
- Deployed a **REST API** coordinating data pipelines and backtesting services for 100+ automated evaluations
- Reworked batch data pipelines for forecasting jobs, improving feature quality and ingestion throughput.
- Optimized **PostgreSQL** schemas, indexes, and caching layers, reducing API latency **18%** under peak load.

Software Engineer Intern

Desung

Jan 2024 – August 2024

New York City, NY

- Engineered behavior-aware game logic in **Unreal Engine**, improving puzzle completion rates by **12%** for users.
- Created internal debugging dashboard for puzzle logic, accelerating **QA testing** and iteration cycles.
- Deployed **RESTful** backend services for multiplayer and analytics, cutting average response time by **120ms**.
- Implemented CI/CD checks for backend services, reducing manual testing before **Docker deployments**.

PROJECTS

Clinical Trial Copilot | *Next.js, TypeScript, FastAPI, Cerebras, Anthropic Claude, PostgreSQL, LangChain, Pandas*

- Built **FastAPI** services for PDF ingestion, structured extraction, and trial metric storage in PostgreSQL.
- Parallelized PDF ingestion and parsing jobs, reducing end-to-end document processing latency by 35% under load.
- Integrated Claude outputs with trial metrics to generate summaries and visual comparisons for **10+** studies.

Seawolf Course Finder | *AWS (EC2, RDS), PyTorch, scikit-learn, BeautifulSoup4*

- Deployed AI course recommender with **TF-IDF + cosine similarity**, semantically ranking **4,000+** courses.
- Hosted course-search API on AWS EC2/RDS, supporting **50K+** monthly queries with logged errors.
- Connected **Flask** API to ML engine and DB layer, serving ranked results in **<1s** end-to-end latency.

GreenPrint | *AWS, Echo (Merit), tRPC, PostgreSQL, Prisma, Express, TypeScript*

- Developed full-stack FinTech app processing **1000+** transactions/month via tRPC + Prisma APIs on PostgreSQL
- Integrated ChatGPT via Echo (Merit) to analyze spend patterns and generate AI-driven **CO₂e** reduction insights
- Optimized **prompt engineering** and token usage for context-rich LLM calls, improving output accuracy 40%

AWARDS AND LEADERSHIP

Business Analytics Club - Co-Founder and Secretary

Hackathon Winner — Stony Brook University HopperHacks (2025)

Datathon Winner (Financial Analysis Track) — Stony Brook Datathon (2025)

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, Bash

Backend: FastAPI, Flask, Express, Node.js, tRPC, REST APIs

Databases: PostgreSQL, Redis, MySQL

Cloud/DevOps: AWS EC2, RDS, S3, Docker, GitHub Actions, Linux

ML/Data: PyTorch, scikit-learn, Pandas, NumPy, LangChain, OpenAI/Anthropic APIs